Impulse response

FIR

$$y(n) = x(n) + x(n-1)$$

n	x(n)	x(n-1)	y(n)
0	1	0	1+0 = 1
1	0	1	0+1 = 1
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0

0.245 − 0.4

0.35×0.7 − 0.8×0.5

IIR

$$y(n) = 0.5x(n-1) + 0.7y(n-1) - 0.8y(n-2)$$

n	x(n)	x(n-1)	y(n)	y(n-1)	y(n-2)
0	1	0	0	0	0
1	0	1	0.5	0	0
2	0	0	0.35	0.5	0
3	0	0	−0.155	0.35	0.5
4	0	0			0.35
5	0	0			
6	0	0			

$$H(s) = \frac{w_c}{s+w_c} = \frac{6283}{s+6283} = \frac{2\pi \times 1000}{s+2\pi \times 1000}$$

$$F_c = 1000\,Hz = 1\,kHz$$

$$2\pi F_c\, e^{-2\pi F_c t} = \frac{1}{6} e^{-t/6}$$

Added after lecture

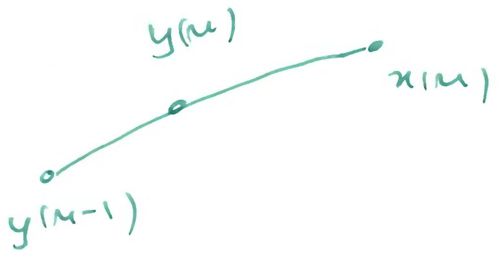
$$H(z) = \frac{w_c}{1 - e^{-w_c T_s} z^{-1}}$$

$$= \frac{6283}{1 - 0.4559 z^{-1}}$$

$F_s = 8000\,Hz$

$\Rightarrow T_s = \frac{1}{F_s} = \frac{1}{8000}\,s$

$e^{-\frac{6283}{8000}} = 0.4559$



BZT

$$\frac{2}{T_s} = 2F_s = 16000$$

$$\Rightarrow F_s = 8000 = 8\,kHz$$

$$\frac{6283}{16000\,\frac{z-1}{z+1}+6283} \times \frac{z+1}{z+1}$$

$$= \frac{6283\,(z+1)}{16000(z-1)+6283(z+1)}$$

$$= \frac{6283\,(z+1)}{(16000+6283)z+(6283-16000)}$$

$$= \frac{6283\,(z+1)}{22283\,z-9717} \quad \begin{array}{l} /22283 \\ /22283 \end{array}$$

$$= \frac{0.282\,z+0.282}{z-0.4361} \quad \begin{array}{l} /z \\ /z \end{array}$$

$$H(z) = \frac{0.282+0.282\,z^{-1}}{1-0.4361\,z^{-1}}$$

$$\frac{Y(z)}{X(z)} = \frac{0.282+0.282\,z^{-1}}{1-0.4361\,z^{-1}}$$

$$= Y(z)(1-0.4361\,z^{-1}) = X(z)(0.282+0.282\,z^{-1})$$

$$y(n)-0.4361\,y(n-1) = 0.282(x(n)+x(n-1))$$